



UNIVERSITY OF CAPE TOWN

TIME SERIES
PORTFOLIO

Insert some title here

Author:
Isabella Lethbridge

Student Number:
LTHISA001

August 20, 2026

Table of contents

Overview	3
Overview	3
Portfolio Structure	3
Reflective Commentary	3
Learning Achievements	3
Methodological Challenges	3
Ethical and Reproducibility Considerations	3
Future Areas for Development	3
AI Use Log	3
Course Section 1	4
Data Statement	4
Teaching Session Summaries	4
Worked Tutorials & Software Implementations	4
Topic Exploration	4
Course Section 2	4
Data Statement	4
Teaching Session Summaries	4
Worked Tutorials & Software Implementations	4
Topic Exploration	4
Substantial Projects	5
Overview	5
Project 1	5
Key Findings	5
Reflections	5
Links	5
Project 2	5
Key Findings	6
Reflections	6
Links	6
Project 3	6
Key Findings	6
Reflections	6
Links	6
Project 4	6
Key Findings	7
Reflections	7
Links	7
Learning Journal	7

Lecture 1	7
Lecture 2	8
Lecture 3	8
Lecture 4	9
Appendix	9
R Code & Scripts	9
Data Access & Repositories	10
Computational Notebooks	10

Overview

Overview

Welcome to my portfolio for the Department of Statistical Sciences. This document showcases my learning progression, statistical understanding, computational applications, and reflections across the course.

Portfolio Structure

- [Reflective Commentary](#) (1000–2000 word synthesis)
- [AI Use Log](#) (Record of generative AI assistance)
- [Course Section 1](#) (Tutorials, model fitting, and reflections)
- [Course Section 2](#) (Tutorials, model fitting, and reflections)
- [Substantial Projects](#) (Four major analytical projects)
- [Learning Journal](#) (Chronological progress tracking)
- [Appendix](#) (Reproducible code and dataset links)[cite: 1]

Reflective Commentary

Learning Achievements

[cite: 1]

Methodological Challenges

[cite: 1]

Ethical and Reproducibility Considerations

[cite: 1]

Future Areas for Development

[cite: 1]

AI Use Log

Date	Tool Used	Prompt / Task Description	Link to Conversation	Verification & Changes Made
YYYY-MM-DD	ChatGPT / Claude	Debugging R code for model fit	Link	Checked output against manual calculations[cite: 1]

Course Section 1

Data Statement

[cite: 1]

Teaching Session Summaries

[cite: 1]

Worked Tutorials & Software Implementations

[cite: 1]

Topic Exploration

[cite: 1]

Course Section 2

Data Statement

[cite: 1]

Teaching Session Summaries

[cite: 1]

Worked Tutorials & Software Implementations

[cite: 1]

Topic Exploration

[cite: 1]

Substantial Projects

Overview

This section contains four substantial projects spanning Section 1 and Section 2[cite: 1].

Project	Section	Project Type	Focus Area	Link
Project 1	Section 1	Simulation Study	Model Comparison	Summary [cite: 1]
Project 2	Section 1	R Package Implementation	Computational Programming	Summary [cite: 1]
Project 3	Section 2	Shiny Application	Interactive Visualization	Summary [cite: 1]
Project 4	Section 2	Exploratory Data Analysis	Real-World Data	Summary [cite: 1]

Project 1

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- [Detailed Analysis & Code Script](#)[cite: 1]

Project 2

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]

- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- [Detailed Analysis & Code Script](#)[cite: 1]

Project 3

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- [Detailed Analysis & Code Script](#)[cite: 1]

Project 4

- **Section & Topic:**[cite: 1]
- **Project Type:** (Implementation / Shiny App / Exploration / Simulation)[cite: 1]
- **Question / Aim:**[cite: 1]
- **Data Used:** (Include source details if real-world data)[cite: 1]
- **Methods Used:**[cite: 1]

Key Findings

[cite: 1]

Reflections

- **What I would do differently next time:**[cite: 1]

Links

- [Detailed Analysis & Code Script](#)[cite: 1]

Learning Journal

This journal documents my continuous learning progress throughout the course[cite: 1].

- [View Chronological Entries](#)[cite: 1]

Lecture 1

Date: 27 July 2026

Topic covered: Foundations

What I Did:

Plotted time series plots for discrete and continuous time series data.

What I Learned:

A time series is an ordered sequence of observations collected over time. The ordering matters since observations are often dependent on previous observations (sequential correlation).

Assumption: equally spaced measurements Assumption: autocorrelation across time

A time series is a ‘single’ measurement taken at multiple points through time to give a sequence of measurements through time.

Noise (stochastic) in data introduces different realisations. An observed time series is only one realisation of an underlying stochastic process. This is important as we are not just trying to model the observations but the process that generated them.

Difficulties/Questions:

Q: How does noise introduce different realisations?

A:

What to Do Next:

????

Links: [Link to Section Work](#) | [Link to AI Log](#)

Lecture 2

Date: 3 August 2026

Topic covered: The Forecasting Problem

What I Did:

????

What I Learned:

A forecast horizon is how far into the future you try to predict values. When looking at time series data we are looking for stable, informative patterns (not just a lot of data).

Forecasting - a prediction about what will happen in the future based on evidence we have (with some uncertainty). This uncertainty gives rise to forecast intervals (not just point estimates)

Goal setting - what we want to happen in the future Planning - how to reach our goal

A random walk is a process where memory is permanent. Every future shock accumulates endlessly resulting in forecast variance growing linearly. A stationary AR is a process where memory decays geometrically. Shocks die out, forcing the forecast back to the process mean.

Difficulties/Questions:

Q: if a process changes over time, at what point can we discard previous observations when forecasting?

A:

What to Do Next:

????

Links: [Link to Section Work](#) | [Link to AI Log](#)

Lecture 3

Date: 17 August 2026

Topic covered: Exploring a Series, Benchmarks and Forecast Accuracy

What I Did:

Worked through *Time Series Analysis in R*

What I Learned:

Difficulties/Questions:

Q: A:

What to Do Next:

????

Links: [Link to Section Work](#) | [Link to AI Log](#)

Lecture 4

Date: 19 August 2026

Topic covered: Three Validation Strategies, Linear Models for the Conditional Mean

What I Did:

Plotted time series plots for discrete and continuous time series data.

What I Learned:

A time series is an ordered sequence of observations collected over time. The ordering matters since observations are often dependent on previous observations (sequential correlation).

Assumption: equally spaced measurements Assumption: autocorrelation across time

A time series is a ‘single’ measurement taken at multiple points through time to give a sequence of measurements through time.

Noise (stochastic) in data introduces different realisations. An observed time series is only one realisation of an underlying stochastic process. This is important as we are not just trying to model the observations but the process that generated them.

Difficulties/Questions:

Q: How does noise introduce different realisations? A:

What to Do Next:

????

Links: [Link to Section Work](#) | [Link to AI Log](#)

Appendix

R Code & Scripts

- `scripts/simulation-study.R`
- `scripts/shiny-app/app.R`

Data Access & Repositories

- Raw Data Files & Data Access Instructions[cite: 1]

Computational Notebooks

- Extended Analysis Reports[cite: 1]